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Advanced Computer Architecture

July 19, 2021

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Name
Last Name
Code Name (Codice Persona)

WRITE YOUR NAME and **CODE NAME**.
Pick 2 out of th 3 problems, **DO NOT DO ALL THE 3**!

Questions (20%)	
Problem 1 (40%)	
Problem 2 (40%)	
Problem 3 (40%)	
Total (100%)	

Question 1

An Embarrassingly Parallel Algorithms is always gaining benefits by an increment in the number of cores provided by the underlying architecture.
Note, in parallel programming, an embarrassingly parallel algorithm is one that requires no communication or dependency between the processes. (True/False)

Answer (Effectively support your answer)

Question 2

Theoretically, an unlimited number of issues will be the best option to achieve the highest performance (True/False)

Answer (Effectively support your answer)

Question 3

Which type of hazards can be caused by the name dependences?

Answer 1: WAR
Answer 2: RAR
Answer 3: WAW
Answer 4: RAW

Answer (list all the correct answer(s) and explain why)

Question 4

In a 5-stage pipeline MIPS architecture, it is possible to forward a data twice from stage-i to stage-j.
As an example, given the following code:

I1 ADDD F2, F4, F6
I2 ADDD F4, F2, F26
I1 ADDD F6, F2, F4

F2 will be forwarder from I1 to both I2 and I3 by using the EX->EX forwarding path (True/False)

Answer (Effectively support your answer)

Question 5

Statically-scheduled superscalar processors are the current state-of-the-art for general purposes processors; some implementations include MIPS and Intel Core i processors (True/False)

Answer (Effectively support your answer)

Problem 1

Assume that the following code is executed on a CPU with TOMASULO and with the following units:

1 LOAD/STORE unit with Latency equal to 1
2 MULT units with latency equal to 10
3 ADD/SUBD unit with latency equal to 2

And the following information:
First load 8 clock cycles (cache miss)
Following loads 1 clock cycle (hit)
R1 = 10

Show the computation steps of the following lines of code (fill the tables for at least 4 clock cycles, not necessary the first 4, not necessary in a row…):

```
Loop:    LD F0 0 R1
         ADDD F6 F4 F0
         MULTD F10 F0 F4
         ADDD F4 F10 F2
         SUBI R1 R1 #10
         BNEZ R1 Loop
```

	Issue	Exec Co.	Write R.

Tables

t = R1 =

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

	Busy	Addr
Load		

F0	F2	F4	F6	F10	F12

t = R1 =

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

	Busy	Addr
Load		

F0	F2	F4	F6	F10	F12

t = R1 =

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

	Busy	Addr
Load		

F0	F2	F4	F6	F10	F12

t = R1 =

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

	Busy	Addr
Load		

F0	F2	F4	F6	F10	F12

t = R1 =

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

	Busy	Addr
Load		

F0	F2	F4	F6	F10	F12

t = R1 =

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
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	Busy	Addr
Load		

F0	F2	F4	F6	F10	F12

t = R1 =

Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
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add3						

	Busy	Addr
Load		

F0	F2	F4	F6	F10	F12

t = R1 =

Name	Op	Vj	Vk	Qj	Qk	Etime
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add3						

	Busy	Addr
Load		

F0	F2	F4	F6	F10	F12

t = R1 =

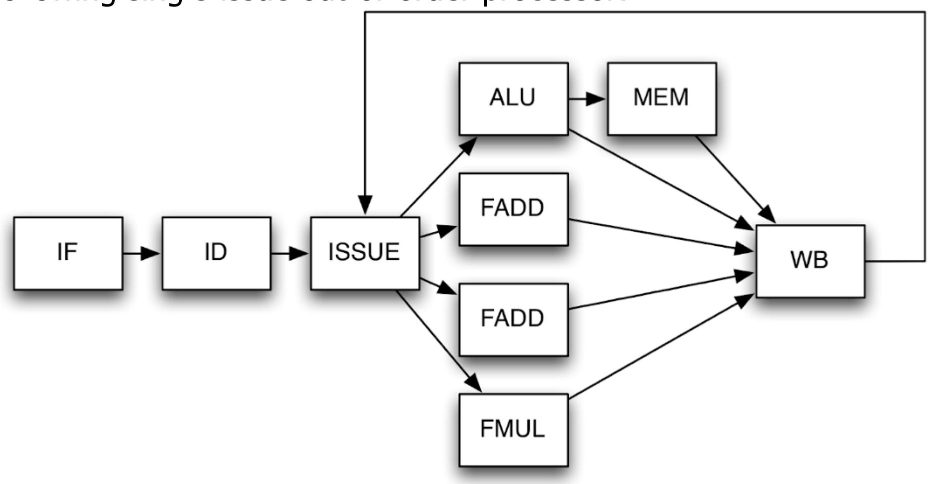
Name	Op	Vj	Vk	Qj	Qk	Etime
mult1						
mult2						
add1						
add2						
add3						

	Busy	Addr
Load		

F0	F2	F4	F6	F10	F12

Problem 2

In this problem we will examine the execution of a code segment on the following single-issue out-of-order processor:



You can assume that:

- All functional units are pipelined
- ALU operations take 1 cycle
- MEM operations take 3 cycles (includes time in ALU)
- FP add instructions take 4 cycles
- FP multiply instructions take 5 cycles
- The WB unit has a single write port.
- RF can write first half of the CC and Read in the second half.
- There is no register renaming
- Instructions are fetched, decoded and <u>issued in order</u>
- The issue stage is a buffer of unlimited length that holds instructions waiting to start execution
- An instruction will only enter the issue stage if it does not cause a WAR or WAW hazard.
- Only one instruction can be issued at a time, and in the case multiple instructions are ready, the oldest one will go first

```
LD     F1, 0(R1)
LD     F2, 0(R1)
ADDD F1, F1, F2
ADDI  R1, R1, 8
MULD F1, F1, F3
SD     F1, 16(R1)
```

1.A List all the possible dependencies/conflicts in the code
1.B Solve considering: WAW and WAR are solved in the <u>FIRST HALF</u> of the WB.
1.C Consider now that the WARs are solved at the <u>END</u> of the ISSUE.

Answer 1.A

Answer 1.B

Instruction	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	C21	C22	C23	C24	C25
I1: LD F1, 0(R1)																									
I2: LD F2, 0(R1)																									
I3: ADDD F1,F1,F2																									
I4: ADDI R1,R1,8																									
I5: MULD F1, F1, F3																									
I6: SD F1, 16(R1)																									

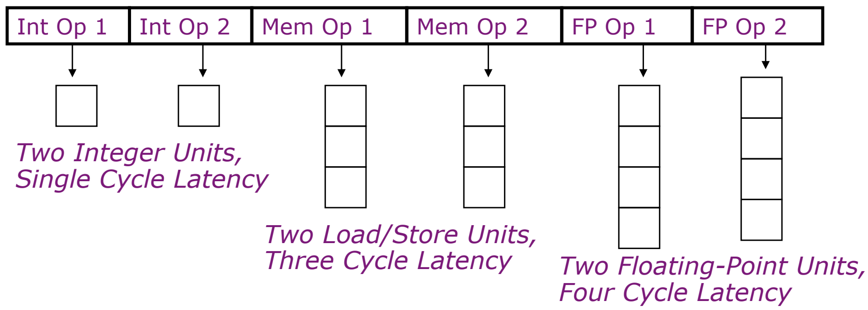
Answer 1.C

Instruction	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18	C19	C20	C21	C22	C23	C24	C25
I1: LD F1, 0(R1)																									
I2: LD F2, 0(R1)																									
I3: ADDD F1,F1,F2																									
I4: ADDI R1,R1,8																									
I5: MULD F1, F1, F3																									
I6: SD F1, 16(R1)																									

Problem 3

P1.A: Explain the main differences between a VLIW architecture and a superscalar architecture, in terms of instructions scheduling, hardware complexity, and code size.

P1.B: Considering the following VLIW “architecture”:



Note that FP Op1 will be just for the FADD, while FP OP 2 is a MULT.
Considering the following portion of assembly, describe the corresponding VLIW code:

```
1. Loop:    LD F0, 0(R1)
            FADD F4, F0, F2
            FMULT F6, F0, F2
            SD F4, 0(R1)
            ADD R1, 8
            BNE R1 R2 Loop
```

How many FLOPS/cycle?

2. With an unroll equal to 4 we can obtain the following code

```
Loop:   LD F0, 0(R1)
        LD F8, 8(R1)
        LD F16, 16(R1)
        LD F24, 24(R1)
        FADD F4, F0, F2
        FMULT F6, F0, F2
        FADD F12, F8, F2
        FMULT F14, F8, F2
        FADD F20, F16, F2
        FMULT F22, F16, F2
        FADD F28, F24, F2
        FMULT F30, F24, F2
        SD F4, 0(R1)
        SD F12, 8(R1)
        ADD R1, 32
        SD F20, -16(R1)
        SD F28, -24(R1)
        BNE R1 R2 Loop
        SD F16, -8(R1)
```

How many FLOPS/cycle?

Answer P1.A

Answer P1.B.1

	Int1	Int 2	M1	M2	FP+	FPx

Answer P1.B.2

	Int1	Int 2	M1	M2	FP+	FPx